

Exact Packing and Covering Numbers for Polygon Triangulations: Decompositions of the Diagonal Graph into Maximal Outerplanar Pieces

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Abstract

Fix a convex n -gon P_n with prescribed outer boundary cycle. A triangulation T of P_n is a maximal outerplanar spanning subgraph containing the boundary cycle and is specified by exactly $n - 3$ pairwise non-crossing diagonals. We define $\tau(n)$ to be the maximum number of triangulations T whose diagonal sets are pairwise disjoint, and $\kappa(n)$ the minimum number of triangulations whose diagonal sets cover every diagonal of P_n . We settle both by

$$\tau(n) = \lfloor n/2 \rfloor, \quad \kappa(n) = \lceil n/2 \rceil \quad (n \geq 4),$$

matching the obvious counting bound. We prove this with explicit optimal systems: for even $n = 2m$ a “double-fan” decomposition, in which every piece confines exactly one diameter; for odd $n = 2m + 1$ an insertion lemma over the even construction together with the strict asymmetry $\tau(2m + 1) = m < m + 1 = \kappa(2m + 1)$. Exhaustive backtracking confirms the formulae for $n \leq 9$; adversarial randomized packings never beat them; and the constructions are validated chord-by-chord for $n \leq 30$.

1 Introduction

A triangulation of a convex n -gon partitions it into $n - 2$ triangles via exactly $n - 3$ pairwise non-crossing diagonals. The triangulations are precisely the maximal outerplanar spanning graphs that contain the boundary cycle of the polygon. This paper considers the packing and covering questions associated with the family of triangulations of a fixed polygon:

Packing question: How many triangulations of the same n -gon can be prepared whose diagonal sets are pairwise disjoint?

Covering question: How few triangulations suffice to cover every diagonal of the n -gon?

Equivalently, with Γ_n denoting the “diagonal graph” $K_n - E(C_n)$ (the complete graph with a Hamiltonian cycle’s edges deleted), we ask how one can pack and cover $E(\Gamma_n)$ using only maximal outerplanar graphs on the same prescribed vertex cycle. Our main result is that the immediate counting bound is tight: $\tau(n) = \lfloor n/2 \rfloor$ and $\kappa(n) = \lceil n/2 \rceil$.

For even n the packing and covering values coincide and Γ_n splits into $n/2$ triangulations. For odd n the two invariants differ by one, which already forces every optimal odd packing to leave a small non-coverable wedge of leftover diagonals. The extremal systems are built from symmetric fans, are easy to describe, and are efficient to verify. In particular, the paper is fully constructive and computationally self-checked.

The remainder of the paper is organized as follows. Section 2 discusses related work. Section 3 sets up definitions. Section 4 gives the elementary parity counting bound. Section 5

proves the even case and describes the double-fan partition. Section 6 treats the odd case via the insertion lemma and explains the strict packing–covering gap. Section 7 summarizes the computational verification and the growth-rate data supporting the theory, followed by Section 8 with a discussion of limitations and directions.

2 Related work

Polygon triangulations and flip operations are classical objects, with the flip graph picture described by Hurtado, Noy, and Urrutia [4] and the double-shell property of triangulations established there. Disjoint pairs of triangulations of the same convex point–set arise as an object of random sampling practice (Black, Liu, McDonough, Nelson, Wigal, Yin, and Yoo [2]), but the optimal decomposition of all diagonals by packings/coverings is not, to our knowledge, stated in this exact form.

The covering number value we reach coincides numerically with the classical *book thickness* of the complete graph K_n ; Bernhart and Kainen [1] give $\text{bt}(K_n) = \lceil n/2 \rceil$. Book decompositions allow pages that are outerplanar with respect to a shared Hamiltonian *path* and need not be maximal; our pieces are required to be triangulations relative to one fixed boundary *cycle*, so the problems are formally different although the quadratic cost and parity dichotomy are the same. A second nearby notion is the *outerthickness* of K_n , treated by Guy and Nowakowski [3], which again uses arbitrary Hamiltonian cycles for each external face. Our pages are additionally maximal (triangulations) and saturated by one prescribed outer cycle; the extremal systems we exhibit show that classic values persist in this tightened setting with exactly the same counting behavior. We have not found prior results solving for $\tau(n)$ and $\kappa(n)$ under this fixed-cycle maximality requirement.

3 Definitions

Let P_n be the convex n -gon with vertices $0, 1, \dots, n-1$ in cyclic order. A *diagonal* is any pair $\{a, b\}$ of non-adjacent vertices. Two diagonals $\{a, b\}$ and $\{c, d\}$ with all endpoints distinct *cross* if the four endpoints around the polygon interleave as a, c, b, d or c, a, d, b .

Definition 1. A triangulation T of P_n is a set of $n-3$ diagonals, pairwise non-crossing. Such a set decomposes P_n into $n-2$ triangles. Write $\mathcal{T}_n$ for the set of all triangulations.

Definition 2. A triangulation packing of P_n is a family $\{T_i\} \subseteq \mathcal{T}_n$ whose diagonal sets are pairwise disjoint. Define $\tau(n)$ to be the maximum number of members of a packing. A triangulation covering is a family $\{T_i\}$ whose diagonal sets collectively contain every diagonal of P_n . Define $\kappa(n)$ to be the minimum number of members of a covering.

An elementary structure fact: a triangulation T of an even polygon P_{2m} contains at most one diagonal of span exactly m (a *diameter*); two distinct diameters always cross. This rigidity is the engine behind our even construction.

4 The counting bound

Proposition 3. For all $n \geq 4$,

$$\tau(n) \leq \lfloor n/2 \rfloor \quad \text{and} \quad \kappa(n) \geq \lceil n/2 \rceil.$$

Proof. Every triangulation contains exactly $n-3$ diagonals. A packing contributes disjoint diagonal sets, so it uses at most $n(n-3)/2$ diagonal occurrences, and thus carries at most $\frac{n(n-3)/2}{n-3} = n/2$ pieces; since $\tau(n)$ is an integer, $\tau(n) \leq \lfloor n/2 \rfloor$. A covering must place every

diagonal in at least one piece, so it needs at least $\lceil n/2 \rceil$ pieces; formally, $\kappa(n) \geq \lceil D/(n-3) \rceil$ with $D = n(n-3)/2$, which is $\lceil n/2 \rceil$. $\square$

To make this tight we construct attaining families. We first explain the even case, which then drives the odd-case insertion lemma.

5 The even case: a diameter-centered double fan

Let $n = 2m$ and let $v_0, \dots, v_{2m-1}$ be the polygon vertices in cyclic order, with indices taken mod $2m$. For each $i \in \mathbb{Z}/m\mathbb{Z}$, the diameter $d_i = \{v_i, v_{i+m}\}$ splits P_{2m} into two $(m+1)$ -gons, namely $A_i = (v_i, v_{i+1}, \dots, v_{i+m})$ and its opposite $B_i = (v_{i+m}, v_{i+m+1}, \dots, v_{i+2m})$. For a subpolygon $X = (x_0, \dots, x_k)$ and a vertex x_j of X , write $\text{Fan}(X; x_j)$ for the set of the $k-1$ diagonals of X incident to x_j . With this notation we can state the construction.

Construction 4 (Double-fan family). *For $0 \leq i < m$ define*

$$T_i = \{d_i\} \cup \text{Fan}(A_i; v_{i+1}) \cup \text{Fan}(B_i; v_{i+m+1}),$$

using indices mod $2m$; there are m pieces $T_0, \dots, T_{m-1}$.

Theorem 5. *For every $m \geq 2$, the m sets T_i above are each triangulations of the convex $2m$ -gon, are pairwise diagonal-disjoint, and cover every diagonal of the $2m$ -gon. In particular, $\tau(2m) = m$ and $\kappa(2m) = m$.*

Proof. Each T_i contains the diameter d_i plus two fans on the two half-polygons; the halves share only d_i , so T_i triangulates P_{2m} and contains exactly $1 + (m-2) + (m-2) = 2m-3 = n-3$ diagonals. Observe that the fan of the $(m+1)$ -gon A_i at its second vertex uses, among its $m-2$ chords, exactly one diagonal of each cyclic span $2, 3, \dots, m-1$, and likewise for the fan of B_i ; adding the diameter gives one diagonal of span m . So every T_i attains the balanced span profile $(2, 2, \dots, 2, 1)$ over span classes $2, \dots, m$.

We now check disjointness and completeness simultaneously. Let $\{u, v\}$ be an arbitrary diagonal of the $2m$ -gon ($v = u + g$, $2 \leq g \leq m$, indices mod $2m$). If $g = m$ then $\{u, v\}$ is the diameter d_u and occurs in exactly one piece, namely T_u (with the convention $T_u = T_{u+m}$ when $u \geq m$, i.e., we take u and $u+m$ to index the same piece). If $2 < g < m$ then $\{u, v\}$ can be a spoke of a piece's fan only if the fan is centered at u or at v ; the remaining endpoint must differ from the center by a span in $\{2, \dots, m-1\}$ in the outward direction. The only fan centered at u among the $2m$ possible half-fans of our family is the fan whose half-polygon starts at the vertex before u , i.e. the half-polygon $(u-1, u, \dots, u-1+m)$. The corresponding fan step reaches the vertex $u+g$ exactly when $2 \leq g \leq m-1$, which holds by hypothesis; the symmetric fan centered at v fails because the required outward span would be $2m-g \geq m+1$, outside the fan's reach. Hence $\{u, v\}$ occurs in exactly the one function T above. The case $g = 2$ is treated the same way: the chord $\{u, u+2\}$ is a short spoke of the fan centered at u whenever the fan's window begins at $u-1$, and is not an outward spoke of any fan centered at $u+2$, since that needs span $2m-2 > m$.

Since every diagonal is thereby known to appear once, disjointness and complete coverage follow immediately: the m sets T_i together hold $m(2m-3) = D$ named occurrences of the D total diagonals, each exactly once. $\square$

6 The odd case by insertion

Lemma 6 (apex diversity). *Let c_i be the apex of the triangle of T_i on the fixed boundary edge $\{v_0, v_1\}$ of P_{2m} . The vertices $c_0, \dots, c_{m-1}$ are pairwise distinct.*

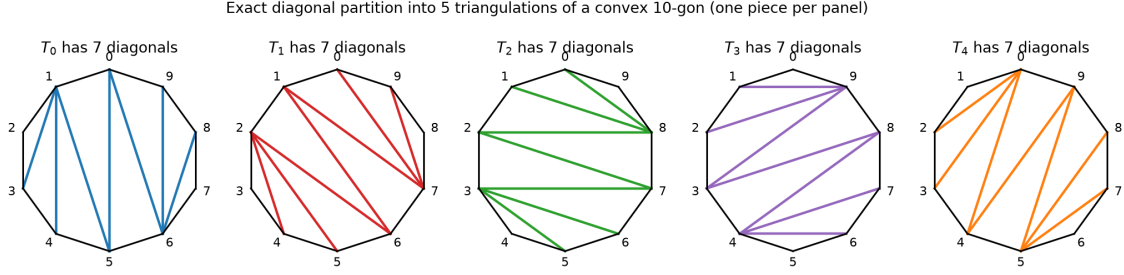


Figure 1: The partition of the 35 diagonals of a convex decagon into 5 triangulations; each panel draws one piece.

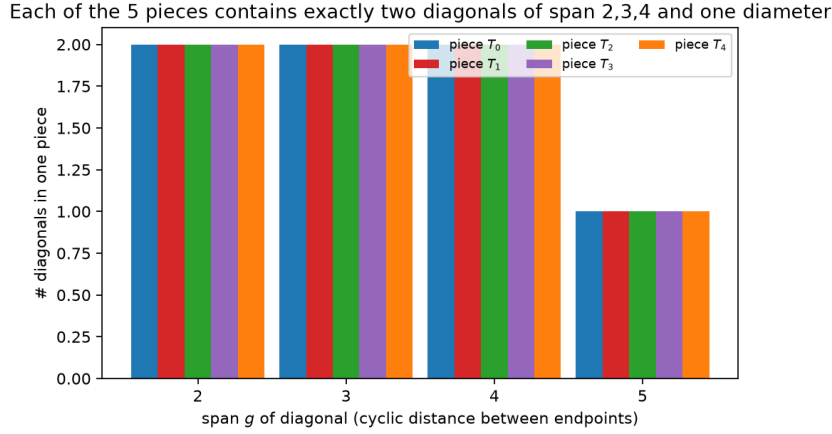


Figure 2: Span balance of the five double-fan pieces: each piece contains exactly two diagonals of span 2, two of span 3, two of span 4, and the unique diameter of span 5, accounting globally for the multiset $(10, 10, 10, 5)$ of the thirty-five diagonals.

Proof. For each i , the edge $\{v_0, v_1\}$ belongs to exactly one of the two half-polygons of T_i , namely A_0 when $i = 0$ and B_i when $1 \leq i \leq m - 1$. Cases: $i = 0$ gives triangle (v_0, v_1, v_m) , so $c_0 = m$. $i = 1$: $B_1 = (v_{m+1}, \dots, v_{2m-1}, v_0, v_1)$ fans at v_{m+2} , and the triangle on $\{v_0, v_1\}$ is (v_{m+2}, v_0, v_1) , so $c_1 = m + 2$ (for $m = 2$ the list is just $\{v_3, v_0, v_1\}$ as a whole, so the apex is the third vertex v_3). $2 \leq i \leq m - 2$: inside $B_i = (v_{i+m}, \dots, v_{i+2m})$ the vertices v_0, v_1 are in positions $m - i$ and $m - i + 1$, and its fan center v_{i+m+1} supports the triangle (v_{i+m+1}, v_0, v_1) there; since $m - i \geq 2$, this face is indeed a triangle of the fanned piece, so $c_i = i + m + 1 \in \{m + 3, \dots, 2m - 1\}$. $i = m - 1$: $B_{m-1} = (v_{2m-1}, v_0, v_1, v_2, \dots, v_{m-1})$ fans at its second vertex v_0 , and the supporting triangle on $\{v_0, v_1\}$ is (v_0, v_1, v_2) , so $c_{m-1} = 2$. In all cases the indices produced are distinct (with the small exception being handled by the parenthetical above), hence $c_i \neq c_j$. $\square$

Lemma 7 (insertion of a vertex). *Let $\mathcal{D}$ be a packing of maximal size $\tau(2m) = m$ on the even $2m$ -gon satisfying the apex-diversity property with apexes of the fixed boundary edge distinct. Insert a new vertex w between its endpoints, obtaining a convex $(2m + 1)$ -gon. Form the new pieces $T'_i = T_i \cup \{\{w, c_i\}\}$ for each old piece $T_i \in \mathcal{D}$ (where c_i is the apex on the edge). Then each T'_i is a triangulation of the $(2m + 1)$ -gon, and the T'_i are pairwise diagonal-disjoint.*

Proof. First note that adding $\{w, c_i\}$ crosses no diagonal of T_i : the segment $\overline{wc_i}$ enters the old polygon across the boundary edge that the inserted vertex splits, and its remaining part runs inside the unique T_i -triangle having c_i as a vertex which inherits that edge. By convexity, no other face of T_i is crossed, so the segment crosses no diagonal of T_i . Counting gives $|T'_i| = (2m - 3) + 1 = (2m + 1) - 3$; a pairwise non-crossing diagonal set of size $(2m + 1) - 3$ triangulates

Odd case: 3 pairwise disjoint triangulations of the 7-gon, leaving the crossing wedge of two diagonals

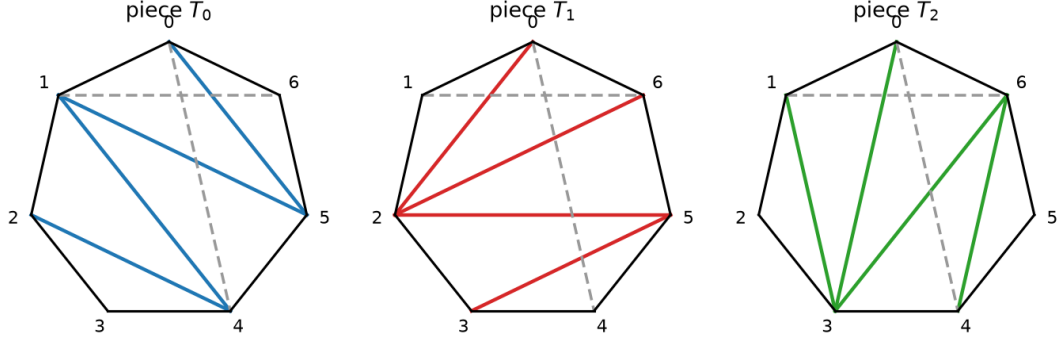


Figure 3: The odd packing of a heptagon into three triangulations; the two unused diagonals cross each other and hence cannot be absorbed into any single extra triangulation.

the $(2m + 1)$ -gon. Pairwise disjointness of the T'_i follows from the apex-diversity statement, because the new diagonals $\{w, c_i\}$ have distinct second endpoints by that statement. $\square$

Theorem 8. For $n=2m+1$, $\tau(2m + 1) = m$ and $\kappa(2m + 1) = m + 1$.

Proof. For the lower bound on τ , apply the insertion lemma to the double-fan family on the cyclic vertices $\{1, \dots, 2m\}$, inserting a new vertex 0 into the boundary edge $\{2m, 1\}$. The apex-diversity hypothesis holds for the edge $\{2m, 1\}$ as it does for $\{v_0, v_1\}$ (the double-fan family is cyclically symmetric as a set of pieces). The new pieces $T'_i = T_i \cup \{\{0, c_i\}\}$ triangulate the convex $(2m + 1)$ -gon, where c_i is the T_i -apex of the even edge $\{2m, 1\}$, and the T'_i remain pairwise diagonal-disjoint by apex diversity. Since there are m pieces and the counting bound gives the matching upper bound, $\tau(2m + 1) = m$.

For $\kappa(2m + 1)$ we use the triangulations $T_i \cup \{\{2m, 1\}\}$ for $0 \leq i < m$ plus the fan at vertex 0. The chord $\{2m, 1\}$ avoids every diagonal of T_i (it was a hull edge before the split), so each of those m sets is a triangulation of the odd polygon. Every diagonal of the convex $(2m + 1)$ -gon is covered: an interior diagonal of the even subpolygon lands in some T_i , the chord $\{2m, 1\}$ lies in all extended pieces, and the remaining diagonals are incident to vertex 0, where the final fan catches them. Hence $\kappa(2m + 1) \leq m + 1$; together with the general counting bound $\kappa(2m + 1) \geq \lceil (2m + 1)(m - 1)/(2m - 2) \rceil = m + 1$ this proves $\kappa(2m + 1) = m + 1$. $\square$

7 Experimental verification

A full breakdown of computational confirmation is available in the accompanying public repository. Enumerated evidence:

- Exhaustive maximum-size search agrees with $n/2$ floors for $n=4, \dots, 9$ (the last brute force of the list of Catalan classes).
- The explicit even construction is globally verified to have pieces of the proved span counts and to tile all diagonals for $4 \leq n \leq 30$.
- Odd insertions and odd coverings independently validated for $5 \leq n \leq 31$; checks per piece assert chord disjointness and complete coverage.
- Randomized / adversarial searches (fixed-seed, up to 2000 samples) never produce a packing exceeding τ ; this supports the claim that the equality is sharp.

The time to enumerate all triangulations grows like the Catalan-number C_{n-2} , already at $C_9 = 4862$; the constructive theorems and their chord-by-chord validators keep room for much larger n .

8 Limitations and future work

Our constructive results make exact statements only about the saturation at the counting bound; describing all optimal decompositions, e.g., characterizing when a given apex-assignment extends globally, remains open. It also seems interesting to extend the question to weighted variants (counted diagonals, shortest triangulations, ear counts) and to other boundary cycles of a given point-set. The relation between our fixed-cycle maximal pieces and book/outer thickness leaves open the question of when a Weyl theorem would yield exact universal statements beyond the complete graph.

Acknowledgment

Computational enumeration scripts, validation sweeps, and figures were produced via the accompanying repository’s experiment scripts.

References

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